

# The information services supermarket — an information network prototype

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*The future of telecommunications is widely accepted as being convergent with that of computing. This future will include a large number of third party service providers offering interactive multimedia services to diverse terminals over a broadband network. A prototype of the service control and management system needed to realise this capability is being built. The experiment has shown that state-of-the-art software techniques such as open distributed processing, artificial intelligence and object orientation must be used and interface standards between networks, terminals, servers and managers are also needed. This is beyond the scope of existing standards, but the designs are based on the deliverables of the TINA consortium, a global telecommunications industry forum which was started in 1993, and which will be influential, it is believed, in future standards.*

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## 1. Introduction

The intelligent network (IN) vision is outlined in the International Telegraph Union (ITU) standards [1] as follows:

*“The objective of IN is to allow the inclusion of additional capabilities to facilitate provisioning of service, independent of the service/network implementation in a multivendor environment. Service implementation independence allows service providers to define their own services independent of service-specific developments by equipment vendors. Network implementation independence allows network operators to allocate functionality and resources within their networks and to efficiently manage their networks independent of network implementation specific developments by equipment vendors.”*

A standard which fully supports this broad vision would certainly satisfy many of the requirements of telecommunications service providers, well into the 21st century. However, by the time the Q.1200 series of IN standards was published in 1992 it was clear that they only supported a small part of the overall vision. It was also clear that further progress within the normal standards process would be very slow.

At the same time it became apparent that there was considerable convergence between the computing industry and the telecommunications industry. The common vision of both the telecommunications operators and the computer

industry was of a future based on information networking. An information network is an infrastructure which:

- facilitates business processes and the efficient use and reuse of information,
- allows diverse businesses to co-operate in whatever way they choose,
- supports user-customisable interactive multimedia applications,
- allows seamless interworking between distributed information sources.

Clearly, this infrastructure will need service implementation independence to be able to deal with the enormous diversity of sources, applications and hosts. It will also need strong standards for interworking — in other words it needs the IN vision to be fully realised.

As part of the long-term IN development process many telecommunications operators had been informally collaborating on this information networking vision through a series of workshops, known as telecommunications information network architecture (TINA) workshops since 1990. At these workshops it was identified that the convergence was both a threat and an opportunity — a threat because it increased the competition and made the need for realisation of the IN vision more urgent; an opportunity because it seemed that the computer industry had some of the answers

and was willing to share its solutions in order to ensure a wide market base.

Managers are thus presented with a triptych of problem, threat and opportunity. One proposal to facilitate the solution of the problems, and thus minimise the threat and make use of the opportunity, was to establish a formal collaboration between the telecommunications operators, the computer vendors and the network equipment vendors which would attempt to bypass the political wrangling associated with formal standardisation and produce a quick *de facto* standard. This collaboration goes by the name of the TINA consortium (TINA-C) [2].

This paper outlines the problems with the IN standards, with respect to the information networking vision, and then introduces TINA-C and its approach to resolving the problems. It then illustrates the use of TINA-C by discussing the implementation of a possible future telecommunications service retailer, and the benefits to the retailer of using the TINA-C approach; this work is being conducted in a project being undertaken at BT Laboratories called ComBAT (common broadband applications testbed).

## 2. The information networking vision

Traditional telecommunications operators' markets are now highly competitive and BT, along with many other telecommunications operators, is responding by seeking new markets and engaging in new activities to serve them. One such market-place is information networking. A selection of information networking services is listed in Table 1.

Table 1 Examples of multimedia information services.

| Interactive services<br>((a) symmetrical duplex<br>communication) | Distribution services<br>(simplex communication) |
|-------------------------------------------------------------------|--------------------------------------------------|
| Multimedia                                                        | Share prices                                     |
| Tele-education                                                    | Weather                                          |
| Tele-shopping                                                     | Electronic newspaper                             |
| Building security                                                 | Electronic publishing                            |
| Traffic monitoring                                                | Tele-advertising                                 |
| LAN-to-LAN interconnect                                           | Tele-education                                   |
| Medical imaging                                                   |                                                  |
| Remote control                                                    |                                                  |
| Computer supported<br>co-operative working                        |                                                  |
| Electronic mail                                                   |                                                  |
| Document transfer                                                 |                                                  |
| Remote control                                                    |                                                  |
| Video on demand                                                   |                                                  |
| Electronic library                                                |                                                  |

Until recently, the principal barriers to mass-market introduction of commercial information services of this

kind have been technological and financial. The Internet, an interconnected user group bound by their interest in exploring, sharing and using multimedia applications, is experiencing explosive growth fuelled by the ready availability of easy-to-use cheap terminals. The range and sophistication of the services available on the Internet is impressive and most of them are free, and many now available via user-friendly hypertext interfaces. It might appear that the principal barriers have been overcome.

However, the services are currently not well commercialised, the user is not sure what to expect from a service, it is not guaranteed to work, a user cannot request exclusive or limited use of the service to improve quality and finding the location of a particular service is by word of mouth. The Internet, as the best known example of a primitive information network, it could be likened to an information bazaar. The buyer has to beware of false claims, overcharging, fly-by-night suppliers, etc, and where successful linking of the user and the desired information can be extremely time consuming.

At present commercial information services, with more guarantees to the user, are starting to appear. Examples include commerce-net (via World Wide Web), Prodigy (supplied by IBM), specialist data providers (e.g. Mead Data) and video-on-demand services (e.g. BT). These services could be compared to an information high street — good services are available but you have to walk all over town to find them and write dozens of cheques.

The ultimate aim is to provide an information hypermarket (or supermarket) which stocks the users' needs on its shelves (whether they are ready-made applications or pick and mix features, own label or third party), provides easy access, has a single billing point and can be seen as a trusted supplier. At current rates of progress this vision will be realised well before the end of the century.

Such an information network will be required to be:

- ubiquitous (to ensure easy access),
- dependable,
- capable of providing user-defined quality of service,
- secure,
- interoperable,
- easy to use,
- responsive,
- anything else a user requires/demands.

To achieve all the above for the complex sophisticated services envisaged it is not enough to provide a broadband network and some simple services that run over it (such as

Freefone and video links). This would be analogous to providing the road network and the fleet of lorries (with cab radios) which are used to ensure the hypermarket is fully stocked. A wide range of management tools and advanced processing techniques are also required. Many of these can be borrowed from the computer industry and built on to the IN infrastructures currently being deployed, but firstly the problems with existing standards must be resolved and interoperability fully understood.

### 3. Problems with IN standards for information networks

#### 3.1 The problem of scope

IN is claimed to be applicable to public switched telephone networks (PSTN), public data networks (PDN), integrated services digital networks (ISDN), and all the services defined for these networks. There are two major difficulties:

- the standards only contain protocols for usage of the PSTN and ISDN and do not address the integration of local access network (LAN) technology or PDN technology,
- the services defined in the standards cover only traditional telecommunications company services such as virtual private networks (VPN), universal personal telecommunications (UPT) and Freefone — they do not currently address the needs of information access services (such as inter-process negotiation on requirements and presentation) at all.

#### 3.2 The problem of interworking

The IN standards attempt to define 'standardised communication between network functions via service independent interfaces'. However, the standards were not originally developed with the intention of supporting interworking between telcos but only between the suppliers to an individual telco. As a result, the standards work well as a procurement template but do not contain enough flexibility to allow inconsistent implementations, which do not interwork (e.g. no global Freefone). This is partly due to a lack of detail in the prescriptions, and partly due to the scope of the definitions, which only attempt to standardise communication between network functions, and do not address communication between network applications. The first problem is being addressed (UPT and global system for mobility (GSM)), but due to associated political sensitivities the solution will consume considerable effort. This means that very little progress is being made on solving the second problem.

#### 3.3 The divergent standards problem

The IN standards cover the domains of network and service management; however the Telecommunications Management Network (TMN) standards [3] (which are derived from the open systems interconnect (OSI) model) deal with management issues in a different way (which is used in ISDN and synchronous digital hierarchy (SDH)). It is not at all clear how these standards are intended to be used together. In addition, new standards such as open distributed processing (ODP), and its derivatives, appear to be likely to increase the confusion further since they assume a radically different origination of software, not used in any existing standards.

#### 3.4 The problem of time

Typically a new standard takes four years to become a stable committee draft and gain approval. The improvements to the IN standards currently under way will be no exception and work on some of the problems has not even started.

## 4. TINA-C

TINA-C is a consortium of approximately 40 telecommunications companies, computer vendors and network equipment vendors, which commenced work in 1993. The telecommunications companies' representatives include such major players as BT, AT&T, Bellcore, NTT, DBP and Cable and Wireless; the network equipment vendors include such major players as Ericsson, GPT and Northern Telecom; the computer vendors include such major players as IBM, HP and DEC. The organisation consists of a core team of 40 researchers, based at Bellcore Labs in New Jersey, USA, and a set of auxiliary projects run independently by the partner companies. TINA-C is defining a *de facto* standard information networking architecture, which will guarantee interoperability between information networks designed using the architecture. The complete architecture will be available some time in 1997. It will be based as much as possible on existing standards and public domain results [2, 4, 5] and will promote:

- universal information access and transport,
- rapid service provisioning,
- tailored services,

- third party development of services and applications,
- service security, reliability and performance.

The process involves the core team describing, the auxiliary projects implementing and criticising, and the core team prescribing on the basis of the results. To date the published results are descriptive but substantial input is being provided by the auxiliary projects which should lead to preliminary prescriptive results early next year (i.e. after two years rather than the usual four for a formal standard). The principal agreements and descriptions are summarised below.

The TINA architecture is based on applying ODP principles of object orientation and distribution to telecommunications system design using TMN managed objects and IN concepts for service management and control.

The TINA-C architecture is based around a core of an object management group [6] distributed processing environment (DPE). This provides for communications between objects, dynamic bindings via a trader function and notification servers to provide management information (e.g. faults, performance). Extensive use is made of autonomous collections of distributed objects referred to as agents [7–9]. The architecture is presented in a set of deliverables which are written in a mixture of English and formal object oriented representations such as Rumbaugh [10] and Guidelines for the Definition of Managed Objects [3].

At the highest level, the TINA-C entities resident on the DPE can be summarised by Fig 1, which describes the various components TINA-C has defined to be necessary to

set up a service connection. The components described are recursive, so, if a service was made up of several other (more basic) services, then each service component could potentially have an associated service session for management of that component.

These TINA components and the interactions between them are now described.

- A user makes some demands (Fig 1, interaction 1) via his set-top box, PC or telephone, i.e. the customer premises equipment. The user can make inputs to impose some personal (or global) preferences either directly (via a keyboard, keypad, swipe card, etc) or indirectly by pointing at preferences for other services.
- Any changes and demands are passed on (Fig 1, interaction 2) to the user's terminal agent (situated locally, but could be held remotely), which can then act on some of the demands (or preferences) and add information to the demands (for instance CPE capabilities).
- The terminal agent then passes on (Fig 1, interaction 3) any information directed to the user agent, along with its additions. The links from the user's CPE to the user agent via the terminal agent is a logical one, the terminal agent merely acting as a filter, adding or removing information as required.
- The agent acting on behalf of the user (the user agent) then makes requests (Fig 1, interaction 4) to the relevant service session manager (say for a video conferencing service) about the requirements that the

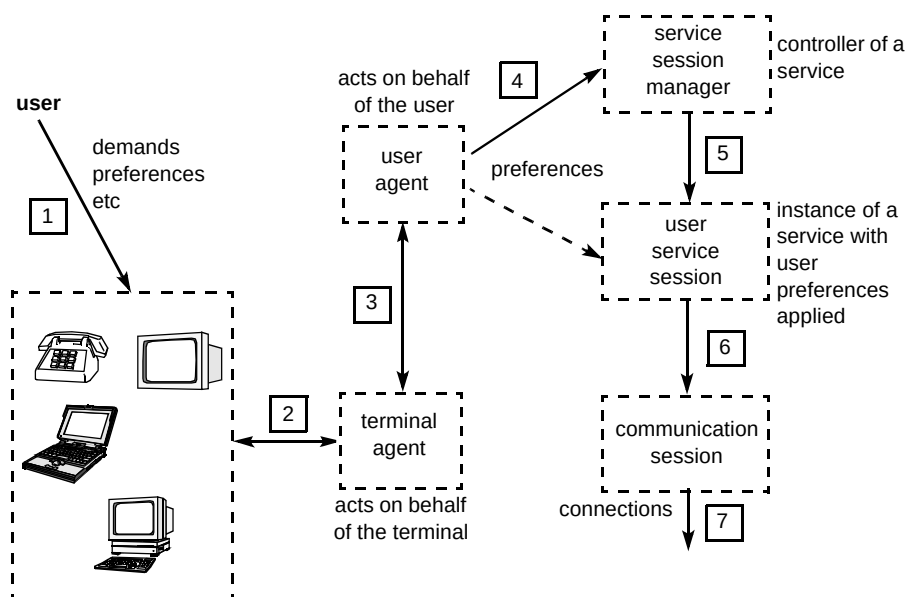


Fig 1 TINA-C access components.

user has directly or indirectly expressed (through preferences communicated via the terminal agent to the user agent).

- These requests are acted on by a factory (Fig 1, interaction 5), which is a component of the service session manager. The factory produces a customised version of the service session, called a user service session. Depending on the constraints imposed by the service session the user may now have some level of control over the service session. For the example of a video conference, the service session class may be set up in such a way as to give chairman privileges to the first entrant into the video conference (or the first user of the video conference service session factory object). The user agent, or the user, now has to inform the other participants in the conference that they are in a conference in progress. In the case of other users, they too will have a measure of customisability and control over the conference session since they will have their own user service session interface. This interface allows them to define the bandwidth of video windows that are not in use (for instance) and define what video streams should go the set-top box, which video or audio streams to store via the PC, etc. They will, however, only have limited control over the general controls of the conference, how to gain the floor, etc.
- The user service session will inherit certain functionality associated with a conference, the need for video and audio streams and a data stream, all potentially duplex and separately manageable. This is achieved by creating (via a factory) objects called connection sessions that are managed by a communication session (Fig 1, interaction 6). This delivers the potential to dynamically configure the streams in a conference, either directly (i.e. choose not to send video) or as part of the service (reduced bandwidth allocation to a

channel that is not directly involved in the conference) or imposed by the user agent (take advantage of low tariff periods, etc) or the terminal agent (terminal capacity constraints).

- The communication session sets up connection sessions (Fig 1, interaction 7) as required (including packet-switched virtual connections for file transfer, etc).

Figure 2 shows how the connections for a video conference can be picked and mixed; in this case the conference is made up of separate objects controlling asynchronous transfer mode (ATM) and ISDN audio and video streams.

## 5. Information services enterprise model

The BT ComBAT project is attempting to realise an optimal implementation of an information networking environment. ComBAT is primarily focused on the service provision and management aspects not covered by network development or application development. To express the domain of interest more clearly an enterprise model (or operations scenario) has been developed. The enterprise model was derived by interpreting information networking as information retailing and attempting to represent the activities of a hardware retailer/distributor [11, 12] in the model. This analogy is the source of the name information service supermarket (or hypermarket). It was felt that basing the model on an existing implementation in this way would enable productive reuse of existing technology and knowledge, and thus minimise the effort involved. Analysis of this enterprise model has led to the view that a distributed object oriented implementation with well-defined open interfaces is needed. Many (but not all) of the concerns are addressed by TINA-C, so TINA-C has been used as the basis of the designs. The model is shown in Fig 3.

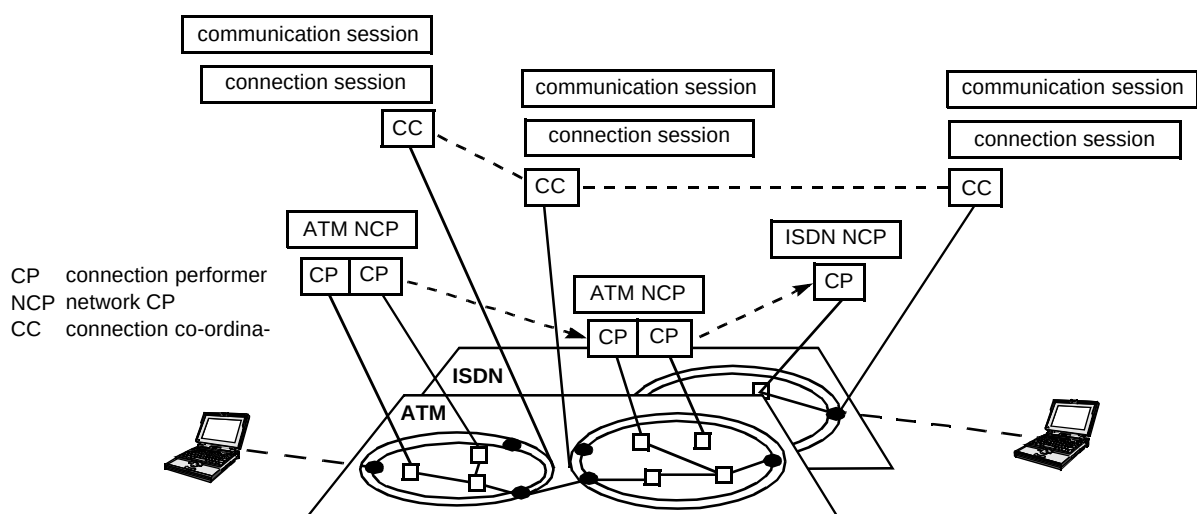


Fig 2 Connection management.

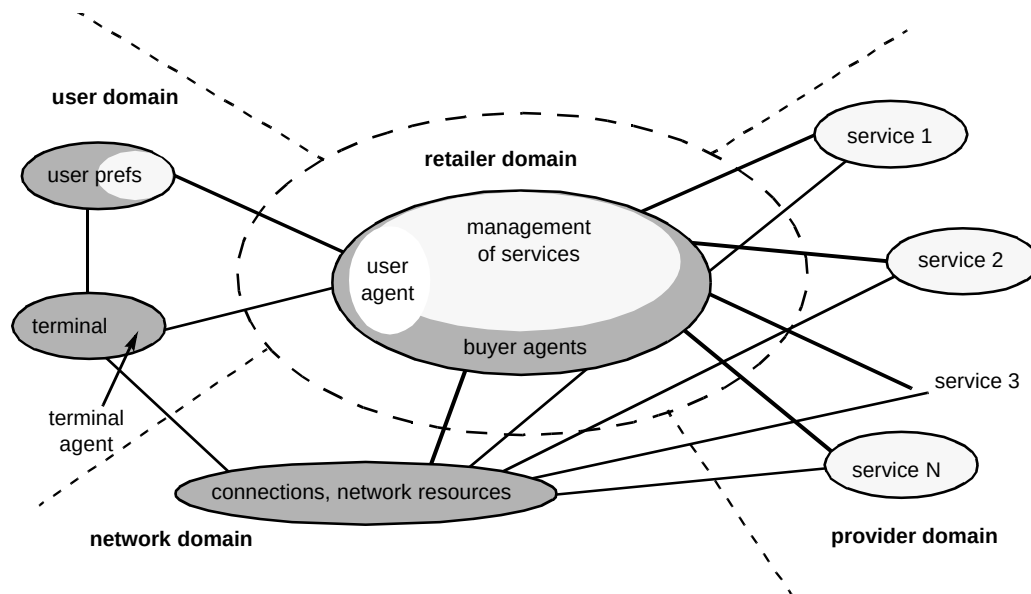


Fig 3 Information services supermarket.

There are four separate areas depicted in the model:

- the user domain,
- the network domain,
- the provider domain,
- the retailer domain.

The user domain represents the domain of interest of users of the services available in the retail domain. It contains the users and items which they own or control such as their personal profiles and applications, and their terminal(s). It also contains a terminal agent which can negotiate with the retail domain when necessary. This agent is intended to include the terminal agent concepts from the TINA-C architecture. It is envisaged that the terminal agent may be supplied by the retailer but owned by the user (or their subscriber).

The network domain represents the domain of interest of the owners of the network used to convey messages from the user to the retailer or other users. It is envisaged that by analogy with 'normal' retailing the user will not have a direct contractual relationship with the network. Instead the network should be regarded as a subcontractor to the retailer. Initial access of the user to the retailer is regarded as being achieved by services such as Freefone which are paid for by the retailer. The network domain is intended to contain the network hardware and associated software, such as connection co-ordinators and managers from the TINA connection management architecture. It also needs to contain connection control features from IN capability set 1/2 (CS-1/2) to enable Freefone and mobility.

The provider domain represents the domain of interest of sources of the services which can be purchased in the retail domain. It contains a set of disparate service providers, each of which provides agents which can negotiate with buyers in the retail domain on supply and payment for services, and similar issues. It is probably a specialisation of the retailer domain with a restricted set of customers and service offerings. This allows the purchase of composite services.

The retailer domain is perhaps the most interesting. It is intended to represent the domain of interest of a retailer who can:

- facilitate access to information and communications services and associated tools,
- act as a middleman or broker for service providers and network providers,
- offer customised, guaranteed services to individual customers,
- manage the services.

It contains a set of intelligent agents which can respond to user and manager demands on an individual basis. These include the TINA-C user agent, service session manager and communications session manager. It also contains a set of management processes which allow managers to:

- monitor activity,
- make configuration changes,
- specify and enforce policies,

- maintain quality, performance, access, etc, as demand and resources vary,
- enforce payment and security related mechanisms,
- assess interactions and add links to third parties,
- advertise services.
- the user domain/retailer domain interface,
- the retailer domain/provider domain interface,
- the retailer domain/network domain interface,
- the user domain/provider domain interface (mediated by the retailer).

A more complete list can be obtained by analysing the activities of existing retailers. It has been found that the above services can be supported by the existing TINA-C resource, fault and configuration concepts, and that the concepts can be reused for high-level designs of security, account and performance management processes. This retailer can be considered almost entirely analogous to the arrangement a supermarket (say a food retail one) has with suppliers of products and the services it offers to those suppliers and the buyers of the products (subscribers). Services (products) are offered by the supermarket on behalf of the service providers to users (end-users that are associated with a subscriber). The service offered can in theory be physically situated anywhere, so it could be resident in the provider's domain and accessed via the supermarket, or it could be resident in the supermarket. In practice the latter is often the most convenient position for it since the cost of large servers supporting many users can be shared between a range of services. On the other hand it may be that the provider wishes to retain the management of his service and merely devolve the access control (thus sharing security costs), or enable his service to be located by an intelligent navigation service offered by the retailer, in which case the server will be in the provider domain.

The model illustrates several important inter-domain interfaces. These include:

In order to support flexible use of services from different providers and connections across different networks, making use of user-domain terminals and applications of diverse types, these interfaces need to be open, i.e. the capabilities required at the interface can be described using a set of defined terms drawn from published standards and agreements. A key concern is to establish the degree of openness required at the inter-domain interfaces and agree how it can be achieved. Refining the descriptive specifications currently provided by TINA-C into a prescriptive standard relies on input from partners focused on the roles they might play in the enterprise model presented earlier. The understanding of the interfaces and their properties gained from the design work, and the degree to which TINA-C enables them to be open, is discussed as an illustrative example in the final part of this paper.

## 6. Supermarket design

As an example, an information service supermarket service access scenario will be used to illustrate a realisation of a system within the model described above. Figure 4 shows a refinement of the enterprise model which illustrates the classes of components used in the design. The scenario contains a single service provider, a single network provider, a single retailer and a set of users.

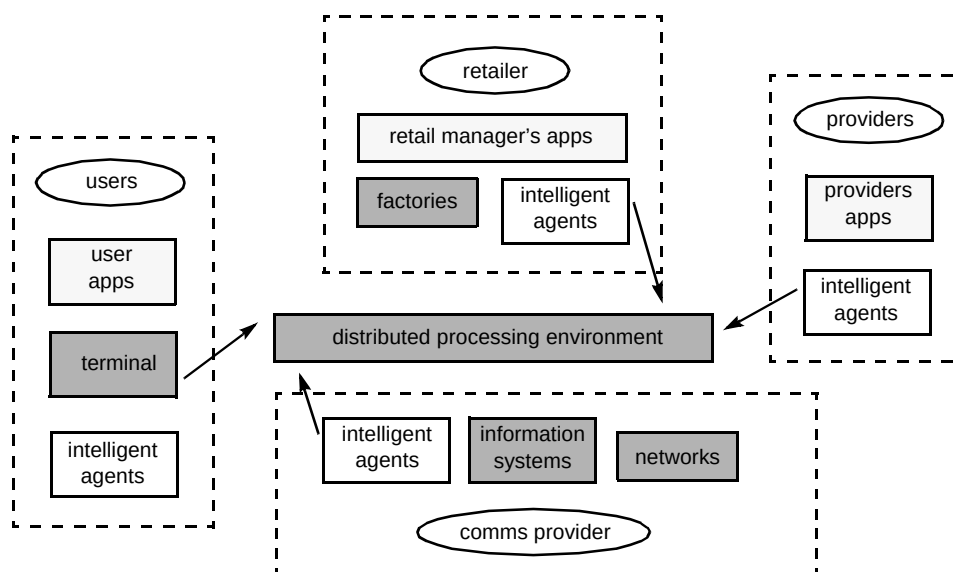


Fig 4 Information network framework.

The provider uses his applications (authoring tools) to create or extend a service. The service is wrapped with some additional code in order to allow it to function in the supermarket (retailer). The wrapper is defined following a negotiation between the agents in the provider domain and the agents in the retailer domain, and its precise form will depend on the relationship between the provider and the retailer. Retailer applications are then used to manage the access of customers of the supermarket to the service. Information about the retailer's customers (subscribers) and users are held by the supermarket in support of, for example, user customisation of services and global user preferences.

The user (no matter what role he is playing) accesses the system via a terminal of some kind (say a personal computer or a set-top box of comparable processing power, with some technology to access the system, DPE kernel, network card, code to access the services on the system, etc). The user initially calls up the supermarket via some kind of Freefone service, which the user has identified through some form of advertising (possibly via the network provider to which the user is connected). On login, authentication takes place; on successful authentication user agents are provided to the user which give them certain capabilities depending on what a subscriber has provided for them.

The agent used will depend on the role that the user is playing. A number of agents and a mechanism for transforming from one role to another have been defined

and designed. The agents and their relationships are depicted in Fig 5.

As can be seen from Fig 5, a number of agents are envisaged, the detailed functions of which are described below along with some other agents not depicted in Fig 5.

- Sales agents — these are the agents the users see when they have logged on to the supermarket and wish to browse and/or try out services. They need to supply subscriber authentication to buy (get access to) services.
- Subscription agent — an agent for the purchase of services for use by end-users. It acts for a number of end-users.
- Management agent — an agent for the management of the service, i.e. service operator role.
- Service provider agent — the access the service provider has to the service provider's service logic.
- Buyer agent — an agent for the purchase of service provider services for resale/inclusion in another service. The actions are split between those by means of which the buyer can act upon the service provider and those that can act on the supermarket.
- Initial agent — the initial agent that is available on start-up of a service for the first time. Control is passed to any of the other agents, although to which one depends on the actions of the user in identifying to the supermarket the role being played.

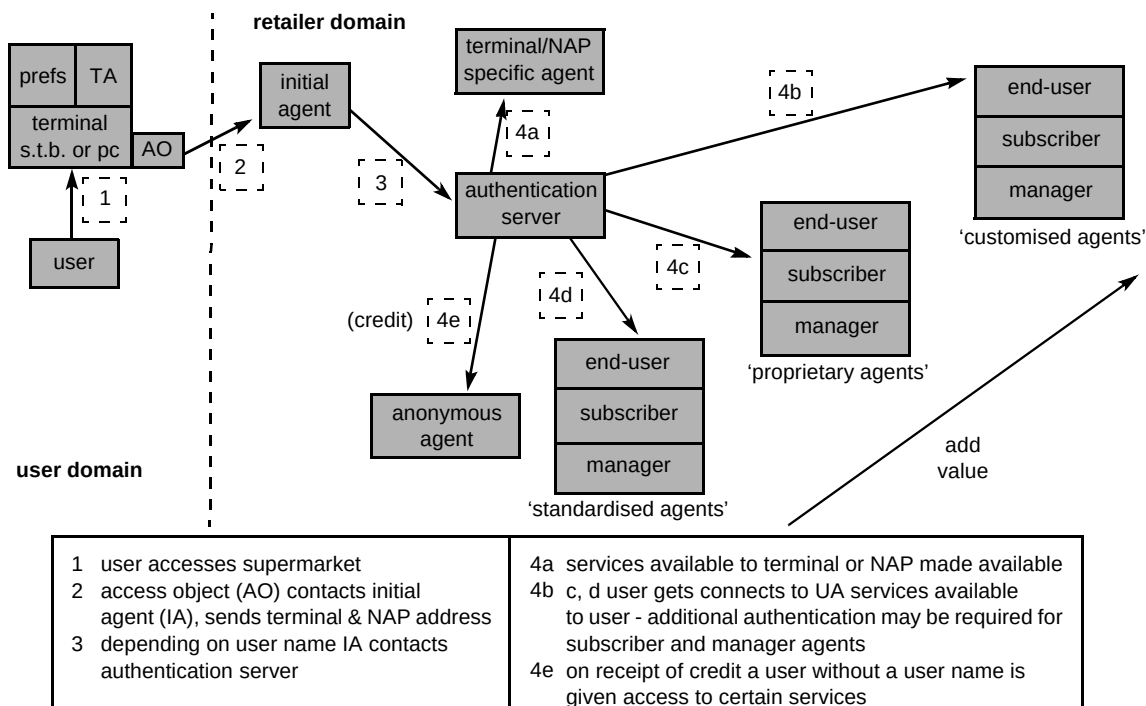


Fig 5 User agents.

- Terminal/NAP agent — services that have been made available to the terminal or network access point (NAP) are used via this agent.
- End-user agent — the end-user aspects. The end-user is not the provider (i.e. subscriber) of the service and therefore cannot browse.

Once the users have obtained their agents they can request any of the services the agents authorise them to access by passing their requests to a factory. The factory first creates a generic service session of the type requested by running the appropriate set of components from a library. The service session then adds any additional user-specific components which are required and becomes a user (specialised) service session. The specialised session then makes use of communications sessions, as in the TINA architecture. Figure 6 illustrates these interactions.

The design is a refinement of the TINA architecture in a number of important respects including:

- the access agents' explicit support for authorisation and role authentication,
- the support for user roles,
- the user service session not necessarily being a standardised component,
- the service session being inclusive of the run-time service session management.

Many of the refinements are now being adopted by the TINA-C core team for inclusion in future deliverables.

Concurrent implementation experiments of this design (in the true spirit of object oriented design) are being performed at BTL. The experiments assumed an ATM network platform since this could provide all the network properties our scenario demanded, without involving substantial network integration tasks. The proprietary signalling scheme supplied by the switch manufacturer was used, since the open ATM standards are not yet mature. Off-line network hardware management (diagnosis, planning, etc) is performed with the assistance of HP Openview. The DPE was based on the OMG compliant Orbix platform supplied by Iona Technologies. A trader was added based on the implementation in Architecture Project Management's ANSAware. So far the implementations have not realised the flexibility inherent in the design since the interfaces are quite complex (see below) and there has not been time to implement all the features necessary. Details of the initial implementation experiments can be found elsewhere [13].

## 7. Inter-domain interfaces

In support of the operations of the supermarket there will be transfers of information across the inter-domain interfaces identified in the enterprise model. The interfaces must allow heterogeneity of applications, terminals, user interfaces, networks and servers. They must also support personalised user preferences, personal and terminal mobility, secure authentication, authorisation and billing, etc. A full definition is clearly going to be complex; for example Table 2 lists many of the capabilities and concepts needed at the interface between the user domain and the retailer domain (i.e. between the terminal agent and the access agent), together with an indication of the wide range of standards which could be used to make the interface open.

It is clear from the table that TINA-C could be the source of many of the open interface concept definitions re-

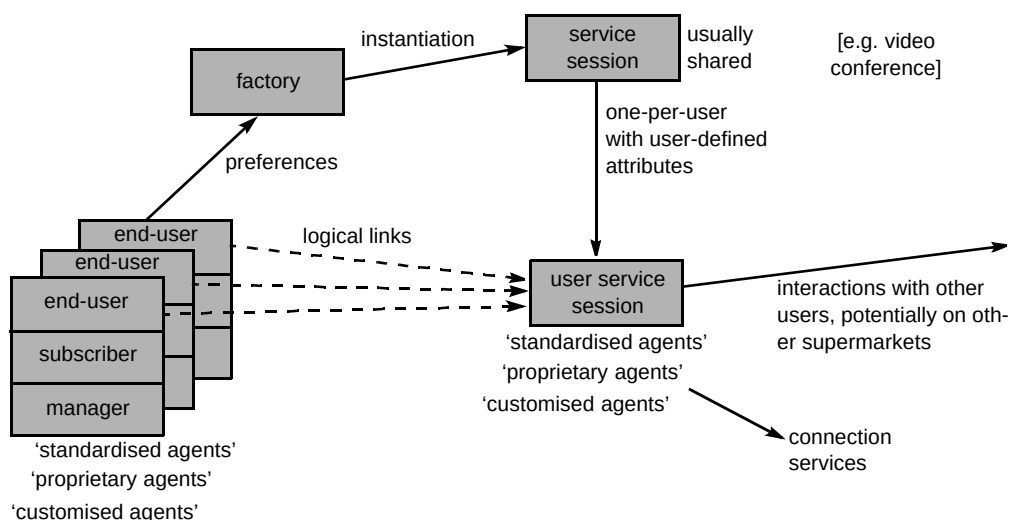


Fig 6 Session interactions.

Table 2 A listing of capabilities at the terminal agent/access agent interface, the concepts which must be defined to support them and some appropriate standards from which the concepts can be drawn.

|                                                 | Network and DPE                                                                                                             |                                                                                                              | Terminals                                                                                                       | Applications                                                                                | User interfaces                                                                                                                                 | Users                                                                                                  |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Required capabilities                           | i) Protocol translations                                                                                                    | ii) Communications between heterogeneous DPEs<br><br>iii) Definition of end points, etc (what is a network)? | Owner defined capabilities<br>Resolve conflicts with user4 preferences<br>Hardware independent O.S. independent | Version and manufacturer compatibility<br>Portable<br>Flexible configuration and deployment | Reproduce user defined look and feel on any device<br>Choose my menus, VR, text styles, colours, speed, and still communicate with capabilities | Set filters<br>Deal with incompatible filters<br>Pick-and-mix features                                 |
| Defined concepts needed to support requirements | i) invocation<br>reply<br>notification<br>argument<br>result<br>object<br>interface<br>stream<br>source<br>sink<br>protocol | ii) bindings<br>channel<br>definition<br>ordering<br>synchronisation<br>transactions<br><br>iii) G.803       | Nucleus<br>Wrappers<br>Capabilities<br>Properties of hardware (machine profiles)                                | Behaviour<br>Dependencies<br>Relationships<br>Entity<br>Format                              | Express<br>— underlying needs<br>— shared attributes<br>— agree capabilities                                                                    | Constraints:<br>time<br>location<br>recipient<br>price<br>interaction<br>QoS<br>permissions<br>details |
| Relevant standards                              | OMG<br>TINA-C?                                                                                                              | OMG<br>IETF<br>TINA-C?<br>OSI                                                                                | DAVIC<br>SPIRIT<br>TINA-C<br>OMG                                                                                | TINA-C<br>Multimedia forum<br>X-Open, OLE                                                   | Nextstep<br>X-11,<br>Microsoft<br>Windows                                                                                                       | TINA-C+<br>auxiliary projects, select from ITU                                                         |

DAVIC The Digital Audio-Visual Council

OSI Open Systems Interconnect

OLE Object Linking and Embedding (Microsoft)

SPIRIT Service Provider Integrated Requirements for Information Technology

IETF Internet Engineering Technical Forum

X,etc Computer Video Display Standards

Nextstep

OO Operating System

quired. These experiments will enable us to steer the TINA-C core team towards prescriptive definitions and accelerate progress to a strong standard. The experiments are also enabling us to provide informed input to a wide range of development programmes.

## 8. Conclusions and recommendations

Many globally aspiring telcos have a strategic business aim of being able to procure interactive multimedia services world-wide in a plug-and-play environment and so give the customer the ultimate in choice. Through an exemplary design, the understanding of the delivery of the advanced features and services currently being discussed has moved on to a stage where the problems appear solvable. A good architecture on which to build solutions and the required open interface standards is TINA-C.

TINA-C was set up in 1993 in an attempt to short cut the standardisation process and build a standard for information networking based around the IN, TMN and ODP standards. To date, the consortium has delivered a set of concepts on which the standard will be based. It has proved possible to create a high-level design of an advanced information ser-

vice and its environment using the TINA concepts, and then successfully implement large parts of that design using commercially available tools. However, major gaps in TINA have been identified, especially for user interface concerns, and much progress needs to be made in the standardisation of key interfaces. In conclusion, while the set of concepts offered by TINA is useful, it needs to be extended to cover these gaps. Further developments should concentrate on the provision of an agreed high-level interface definition language which should be thought of as a client of the OMG interface definition language.

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